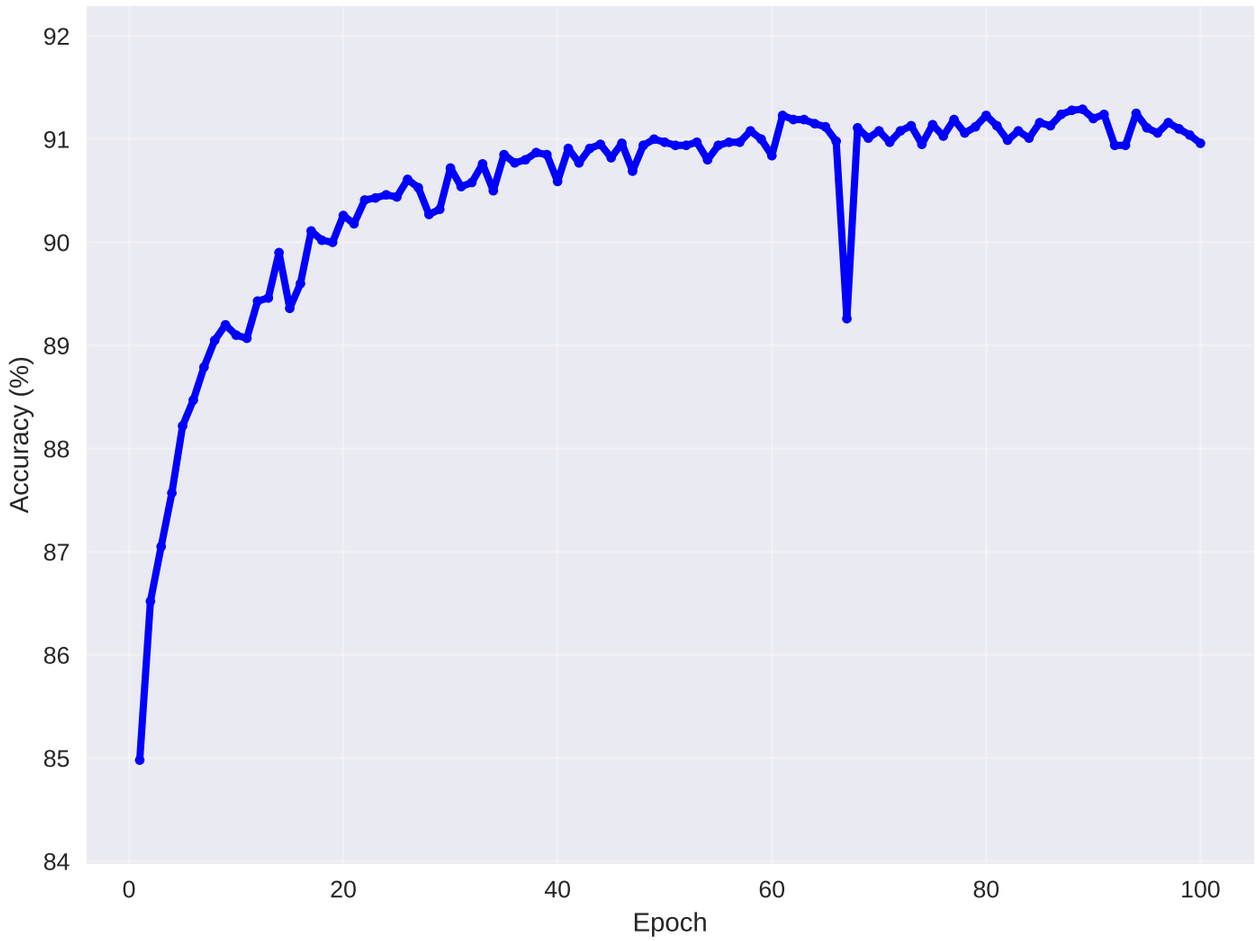
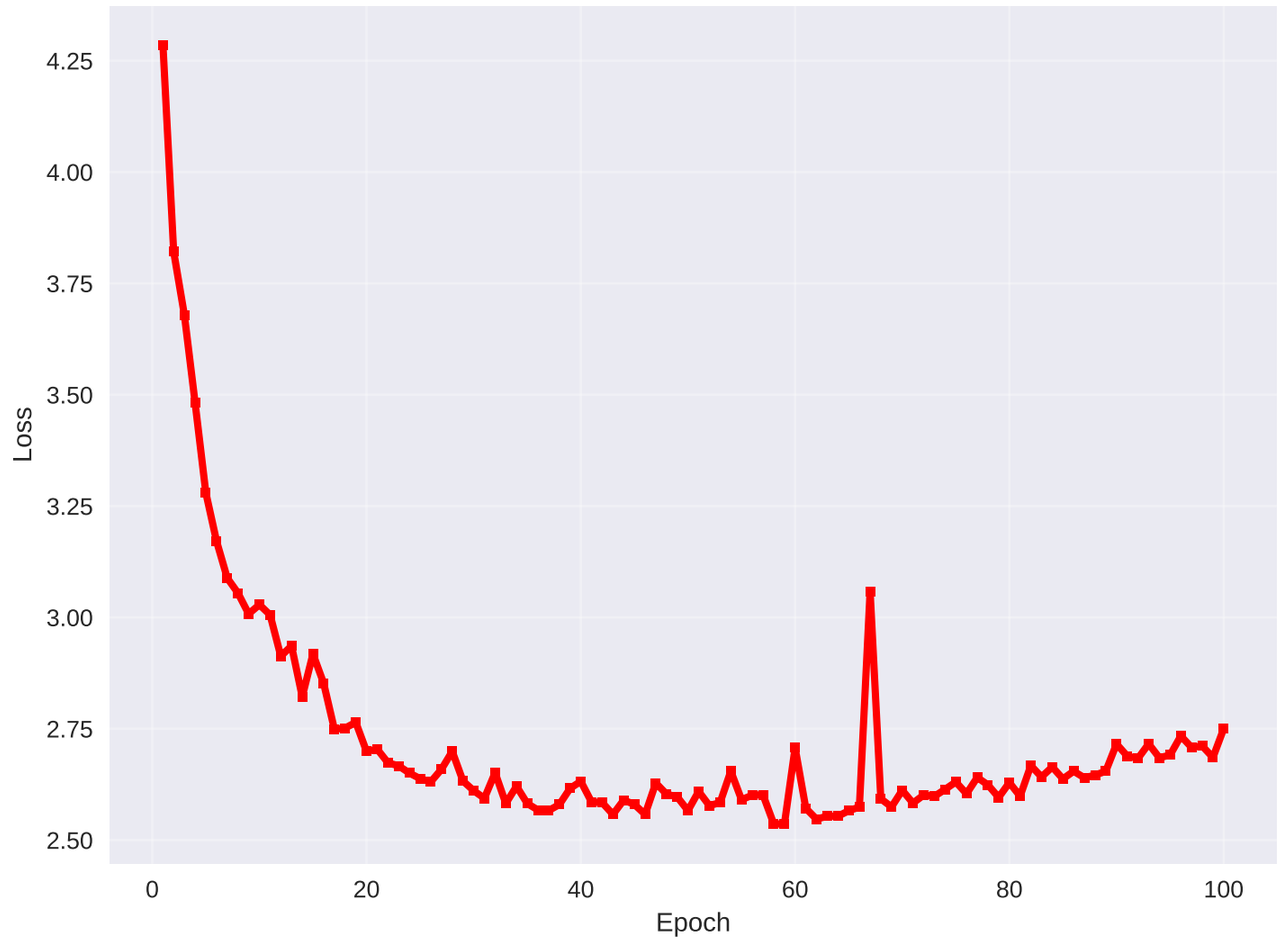
Federated Learning - Main Metrics Overview

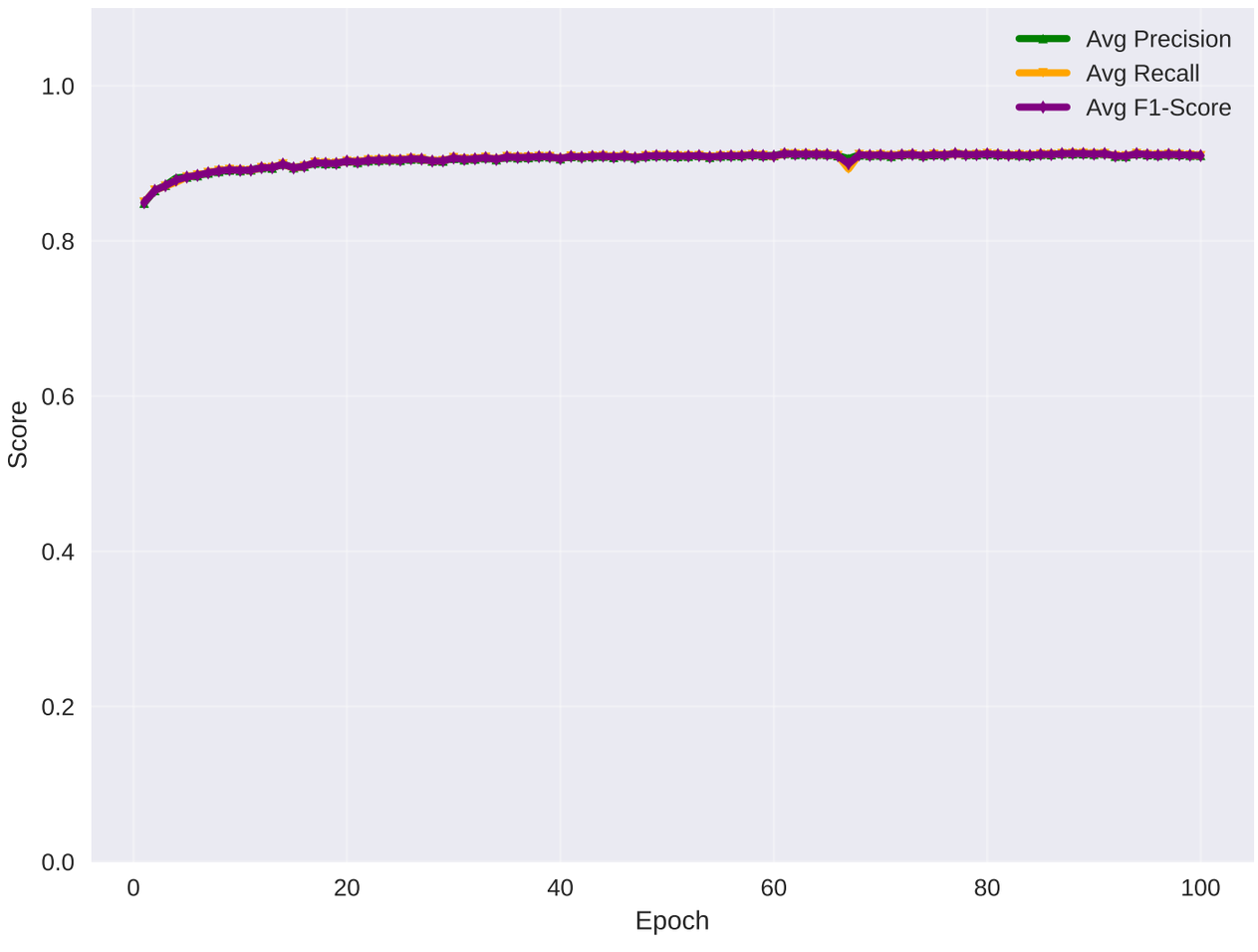
 Global Model Accuracy Evolution



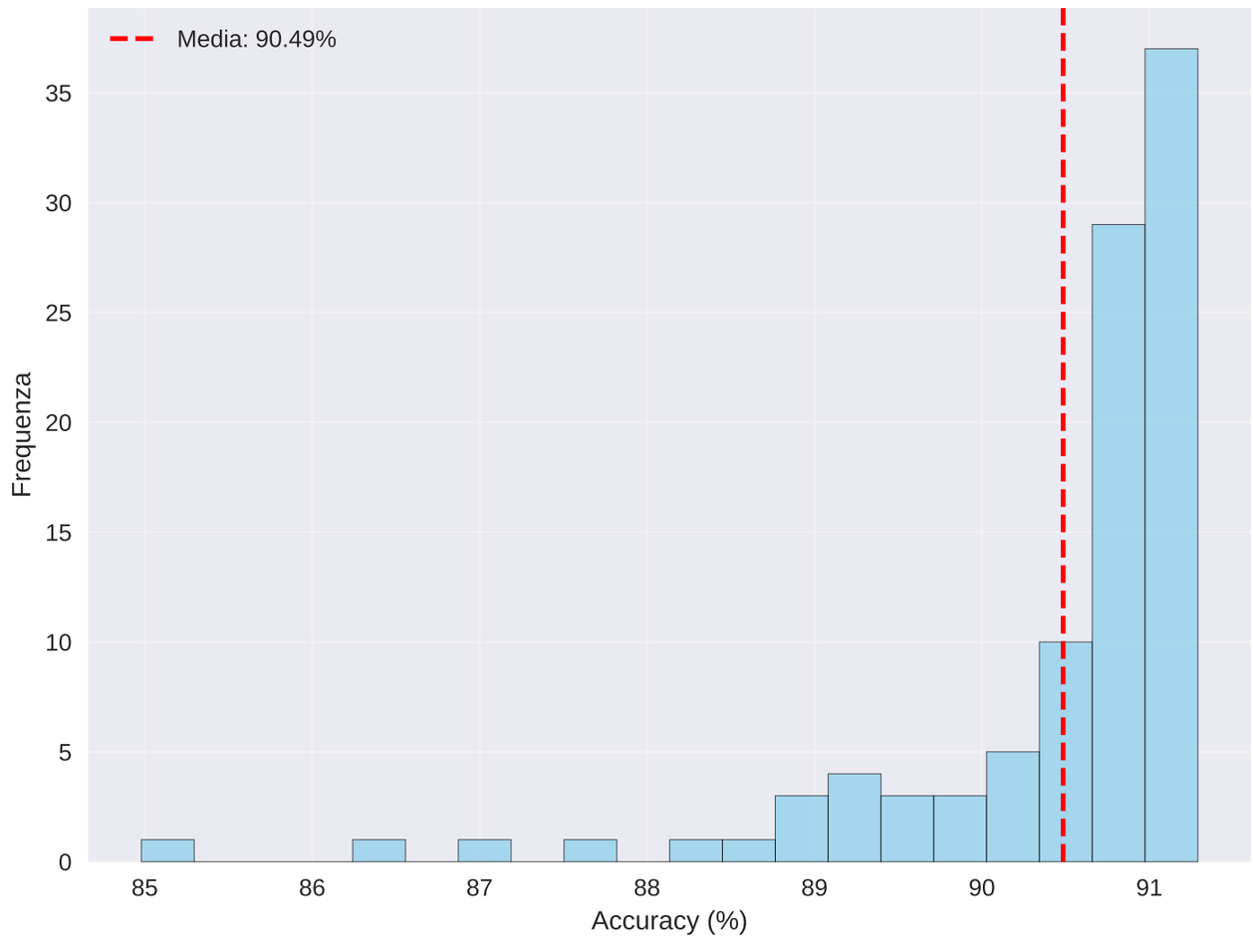
 Global Model Loss Evolution



 Average Precision, Recall & F1-Score

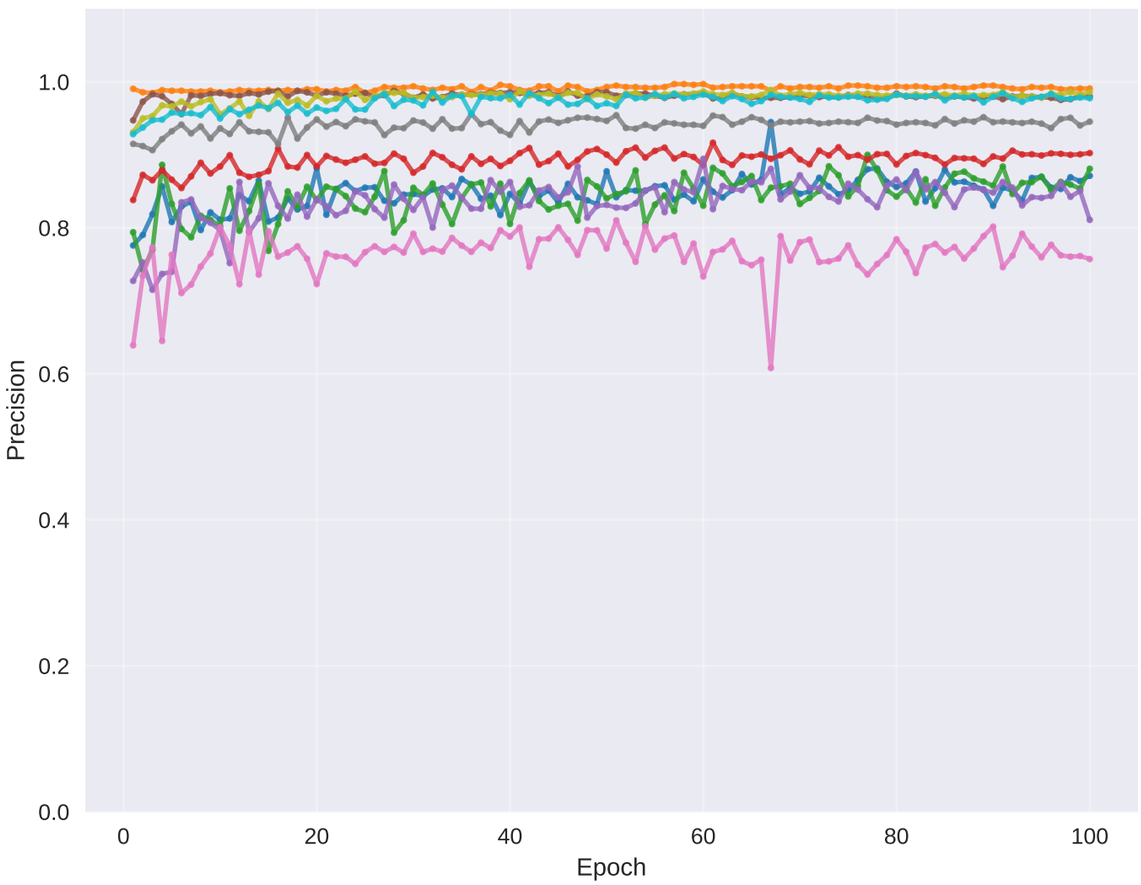


 Distribuzione Accuracy

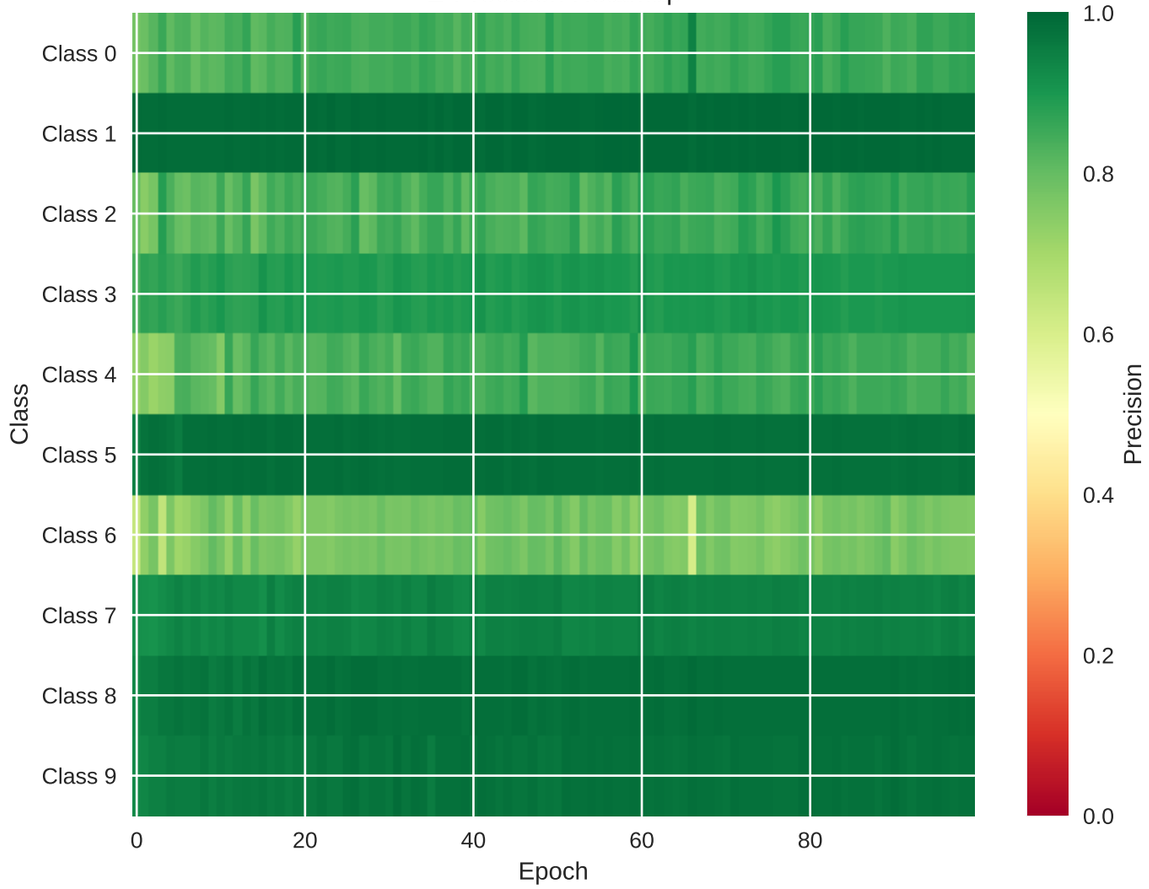


Per-Class Precision Analysis

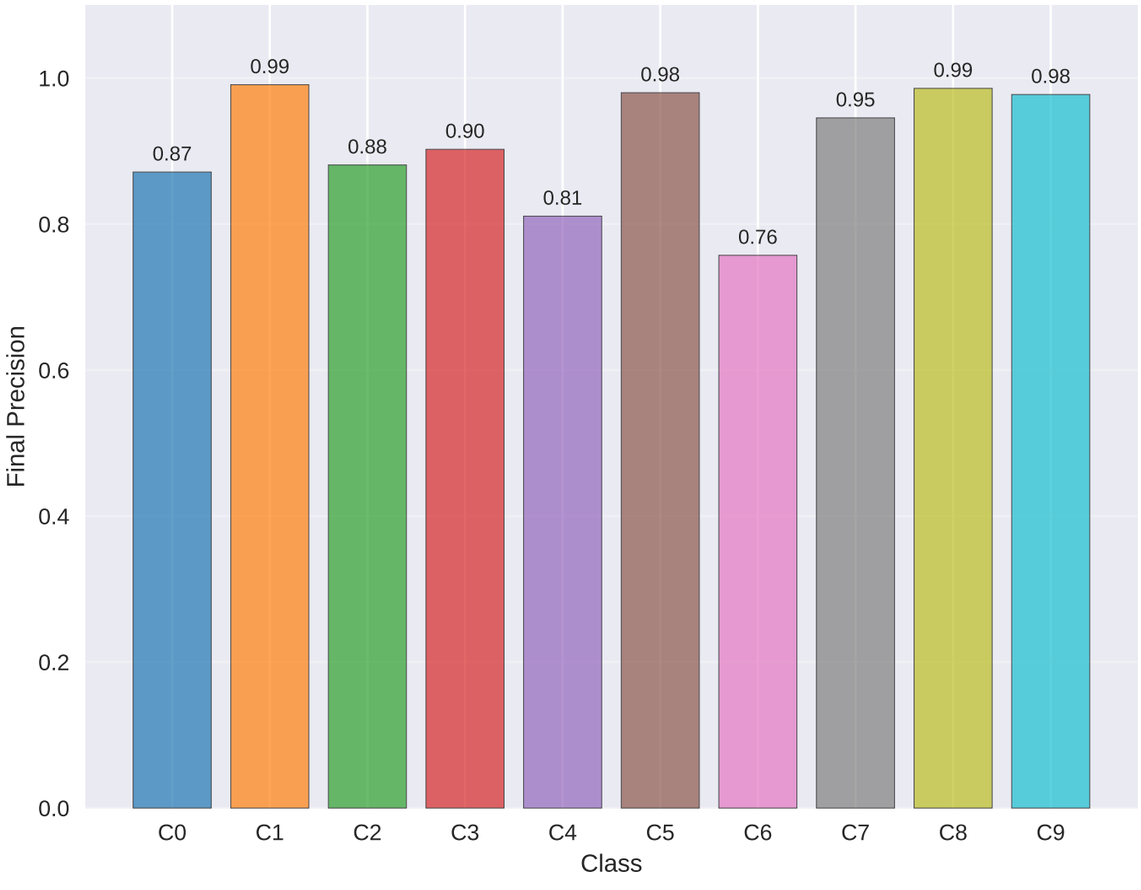
□ Per-Class Precision Evolution



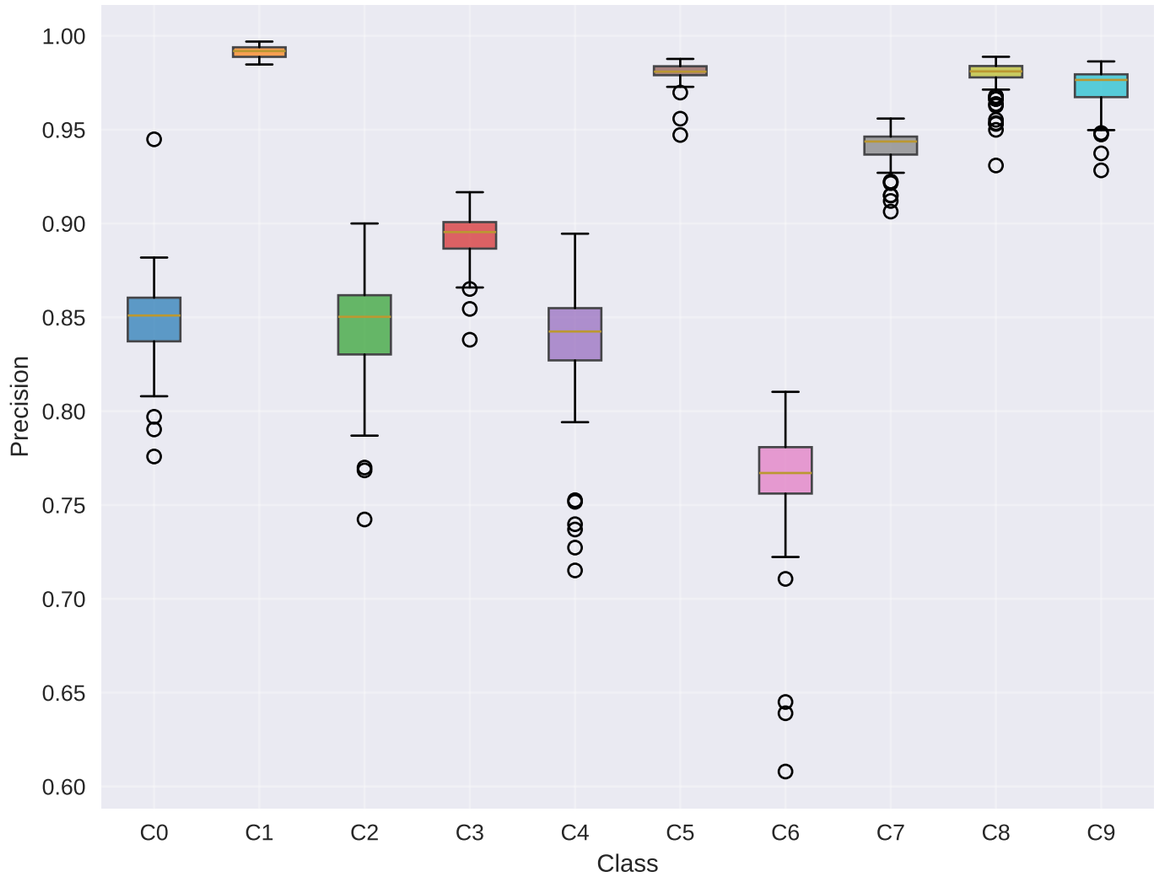
□ Precision Heatmap



□ Final Precision per Class

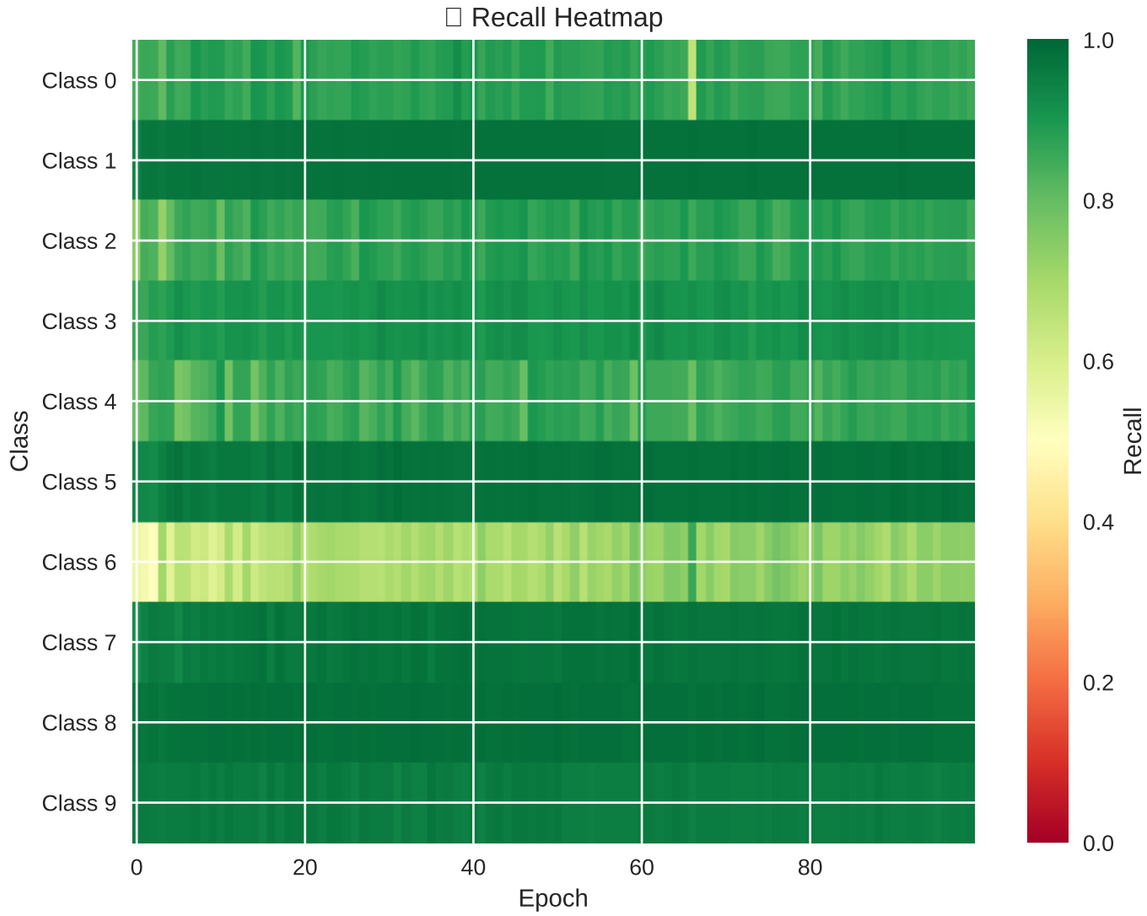


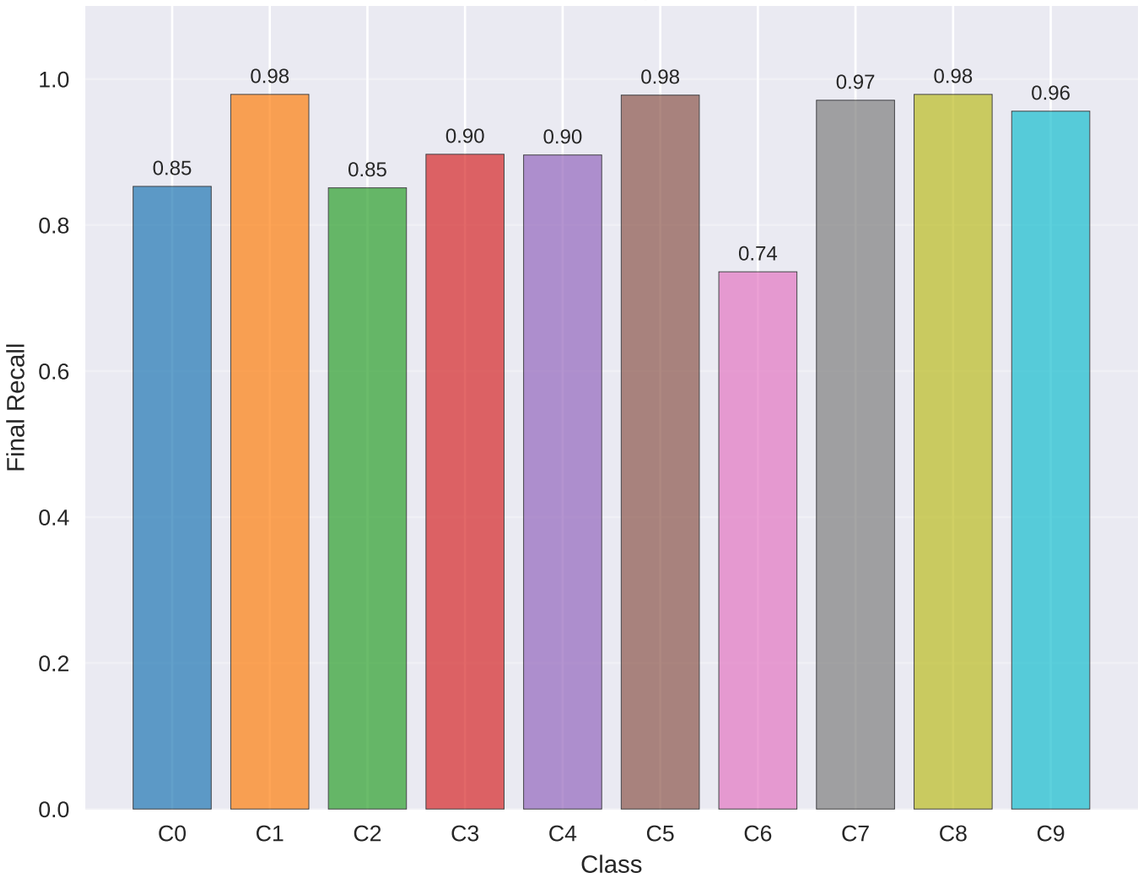
□ Precision Distribution per Class

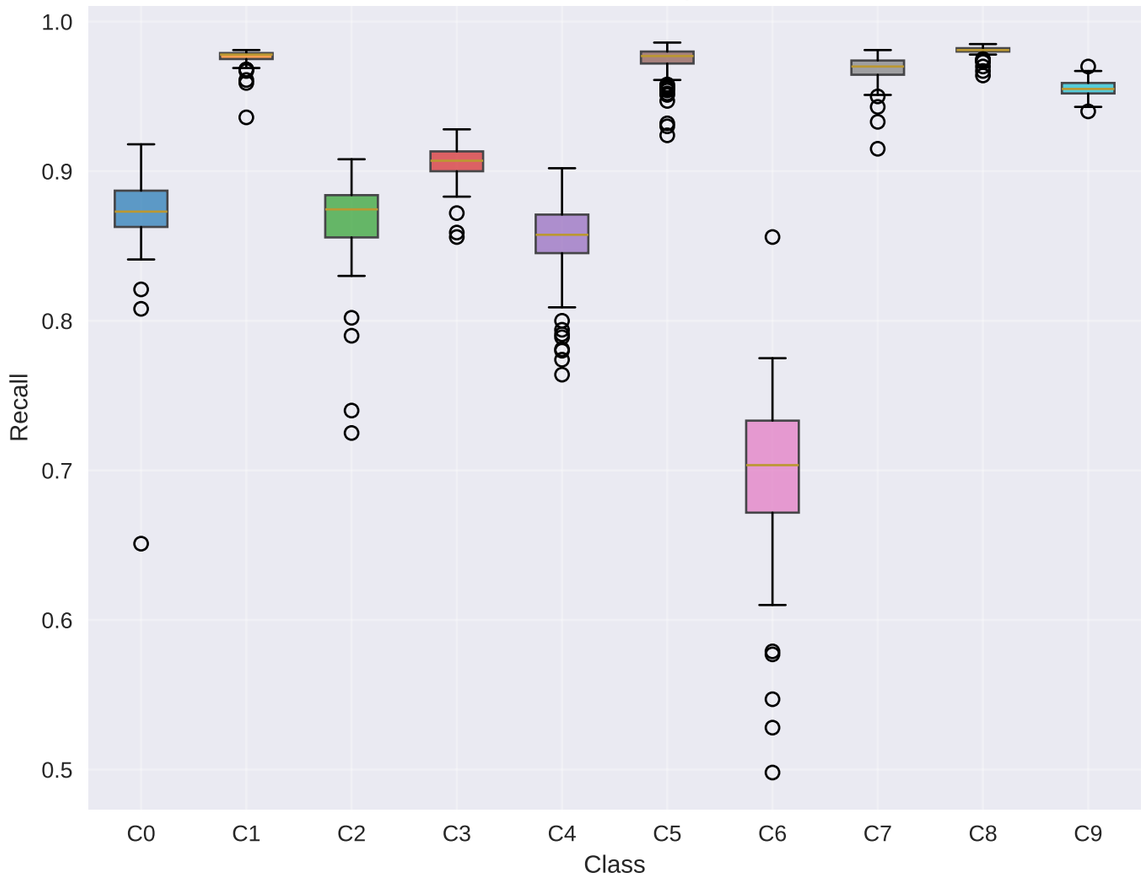


Per-Class Recall Analysis

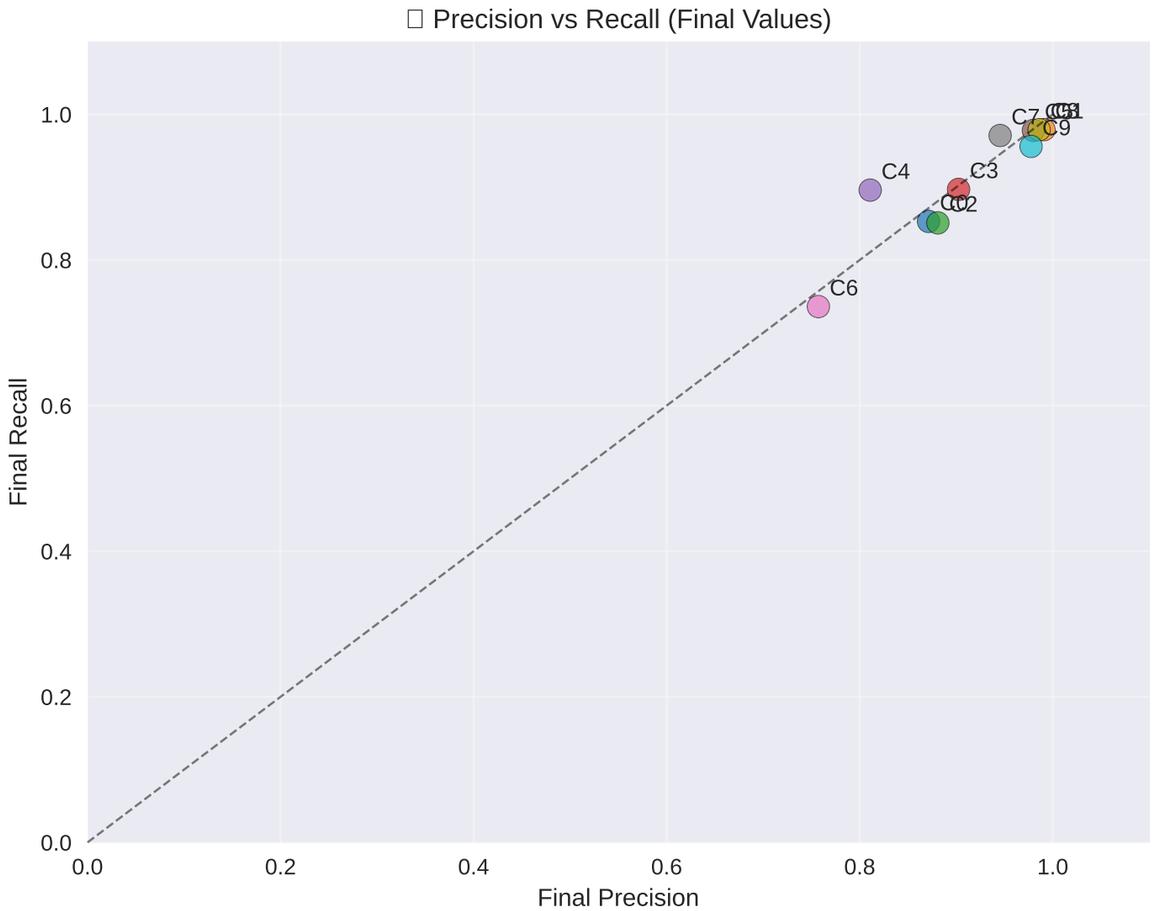




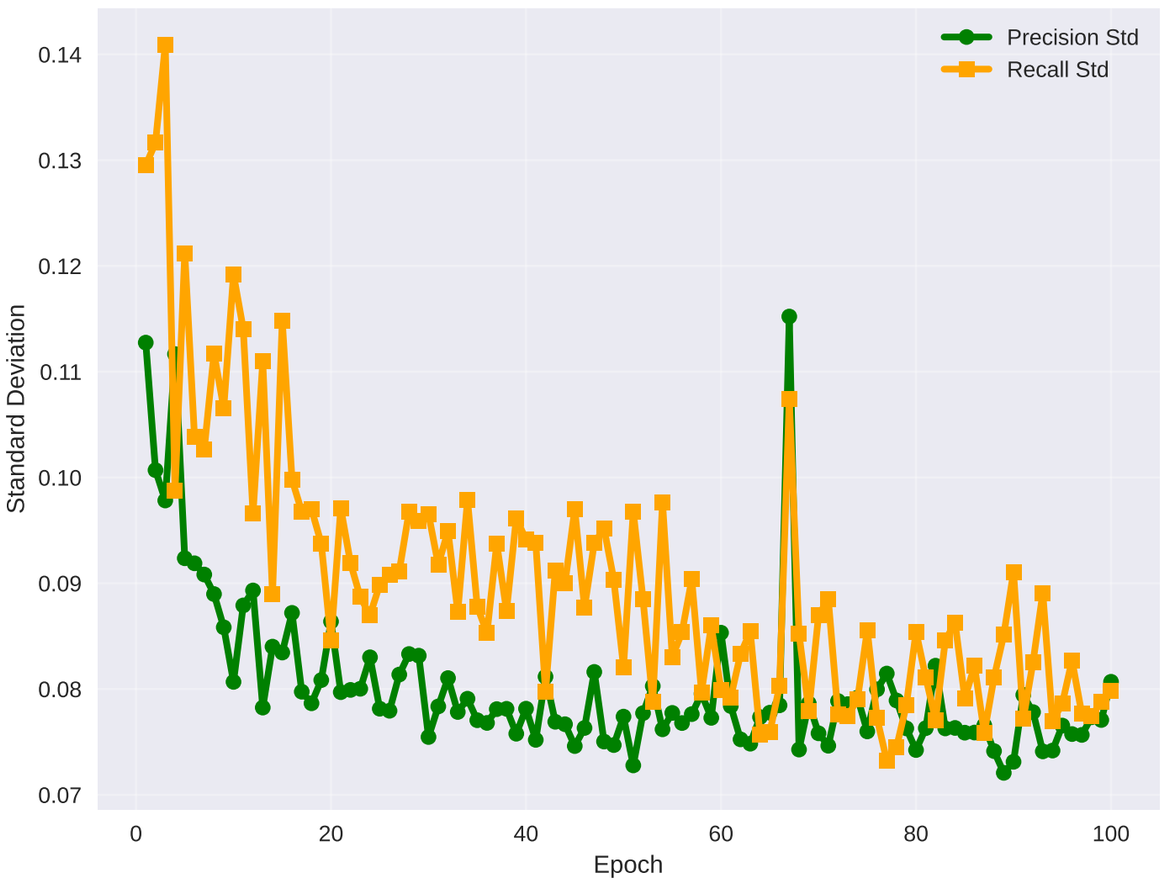


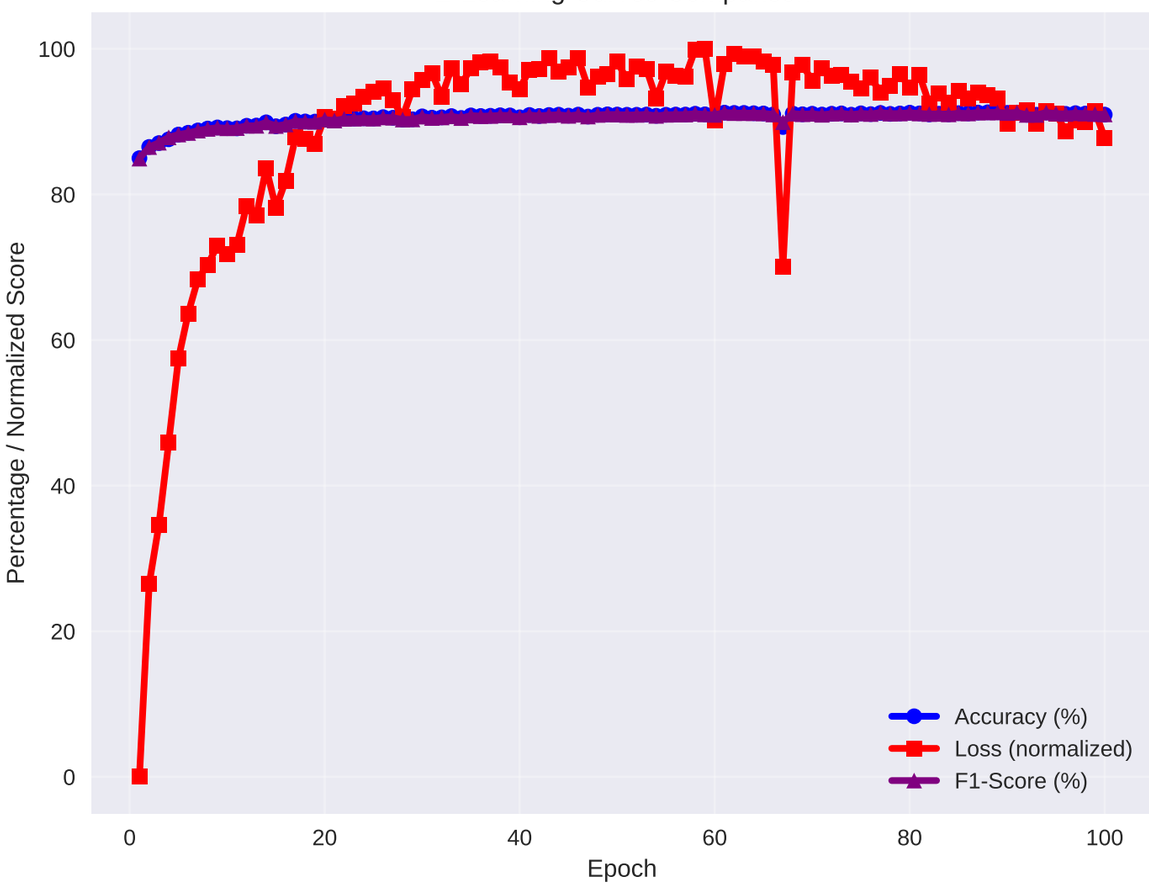


Comparative Analysis









Performance Summary Report

```
FEDERATED LEARNING - PERFORMANCE SUMMARY
========================================

FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-31 20:18:53
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 90.96%
• Loss: 2.7501
• Average Precision: 0.910
• Average Recall: 0.910
• Average F1-Score: 0.910

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +5.98%
• Riduzione Loss: +1.5351
• Stabilità Accuracy (CV): 0.0118

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.991)
• Peggiore Precision: Classe 6 (0.757)
• Migliore Recall: Classe 1 (0.979)
• Peggiore Recall: Classe 6 (0.736)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.871, R=0.853, F1=0.862
• Classe 1: P=0.991, R=0.979, F1=0.985
• Classe 2: P=0.881, R=0.851, F1=0.866
• Classe 3: P=0.902, R=0.897, F1=0.900
• Classe 4: P=0.811, R=0.896, F1=0.851
• Classe 5: P=0.980, R=0.978, F1=0.979
• Classe 6: P=0.757, R=0.736, F1=0.746
• Classe 7: P=0.945, R=0.971, F1=0.958
• Classe 8: P=0.986, R=0.979, F1=0.982
• Classe 9: P=0.978, R=0.956, F1=0.967

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
```